Secondary School Portal

Product Requirements Document (PRD) — Version 1.0

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1. Executive Summary & Goals

The Secondary School Portal is a unified web-based management platform designed to digitize and streamline administrative, academic, and financial operations. By offering tailored dashboards for administrators, teachers, and students, the platform replaces manual workflows with secure, real-time data handling.

Key strategic goals for Version 1.0 include:

- Digitize standard academic operations including enrollment, grading, and attendance tracking.
- Enforce strict Role-Based Access Control (RBAC) and resource-level authorization boundaries.
- Ensure teachers can access only their assigned classes and subjects.
- Ensure students have exclusive secure access to their own academic and financial records.
- Incorporate a dependable payment workflow for school fees with modern gateway integration.

2. Role Definitions

The platform enforces three primary roles, each restricted to its specific operational context:

1. ADMIN: Central authority responsible for system configuration, user provisioning, academic structures, global fees setups, general notifications, and final approval/publication of results.
2. TEACHER: Classroom manager restricted to assigned classes and subjects. Authorized to log daily student attendance, enter continuous assessment (CA) and exam scores, save raw drafts, and submit finalized scores to the administration.
3. STUDENT: End-consumer of school services. Authorized to view personalized dashboards, read global announcements, track outstanding fee balances, make digital payments, and access approved/published academic results.

3. Portal Specifications

3.1 Student Portal

Students interact with a highly personalized, read-only interface focused on tracking progress and managing obligations:

- Dashboard: Displays current class, active academic session/term, fee payment status alert, recent school announcements, and notification of new published results.
- Profile: Centralized view of personal information, enrollment details, admission numbers, assigned class, and guardian contact profiles.

- Results: Secure interface rendering CA, exams, total scores, final grades, and teacher remarks. Crucially, raw data is never exposed; only approved and officially published results are visible.
- School Fees & Payment: Provides a line-item breakdown of fee structures, balances, historical payment transactions, and PDF receipts, alongside a direct checkout button for online payment.

3.2 Admin & Teacher Portals

Administrators utilize a full-suite dashboard to oversee user creation, academic calendar terms, student assignments, and financial records. Teachers access a streamlined interface containing structural elements only for classes and subjects officially assigned to them by the Admin.

4. System Workflows

To guarantee process compliance and auditability, the platform enforces specific multi-step workflows for critical actions:

1. User Management: An authorized Administrator generates a unique user record, configures specific profile attributes, and assigns a singular system role (Admin, Teacher, or Student).
2. Result Submission & Approval: Teacher enters continuous assessments and exam scores → saves dynamic drafts → submits to Admin → Admin reviews → Admin approves or rejects → Admin publishes approved grades → Student gains instant view access.
3. Attendance Logging: Authorized Teacher selects assigned class and subject → chooses date → registers daily presence/absence records → commits data to persistence layer.
4. Payment & Reconciliation: Student initializes a fee transaction → checkout routes to the external gateway → system processes payment → secure Webhook fires → system verifies webhook validity → platform updates user balance and updates payment history ledger.

5. Authorization Matrix

The matrix below outlines the strictly enforced access controls based on roles, physical ownership, and pre-assigned academic contexts:

Resource Description	Admin Access	Teacher Access	Student Access
User Profiles & Configurations	Full Read/Write	Self Profile Only	Self Profile Only
Academic Setup (Classes, Subjects)	Full Read/Write	View Assigned Only	No Access
Draft Academic Results	Read Only	Write Assigned Only	No Access
Published Academic Results	Full Read/Write	Read Assigned Only	Self Results Only
Fee Structures & Invoicing	Full Read/Write	No Access	Self Records Only

6. Minimum Viable Product (MVP) Scope

To ensure rapid deployment and core security, the primary launch scope focuses strictly on the following:

- Authentication: Secure stateless JWT architecture protecting all API routing.
- User Management: Basic administration utilities for provisioning verified roles.
- Academic Framework: Entities to control sessions, terms, classes, enrollments, and teaching paths.
- Grading System: Standard CA/Exam score ingestion, submission pipeline, and approval actions.
- Financial Gateways: Integration with a trusted payment gateway alongside robust idempotent webhook processing to prevent double-charging.

7. Future Horizons & Extensions

Following stable deployment of the MVP specifications, subsequent upgrade phases will introduce extended capabilities:

- Performance Optimization: Distributed caching utilizing Redis instances.
- File Storage: Object storage via MinIO cluster instances for academic document attachments and user avatars.
- Asynchronous Workers: Queue-based worker systems for background notifications and bulk reporting pipelines.
- Interactive Modules: Complete lesson timetabling, online student assignment collection modules, and comprehensive analytics dashboards.
- Extended Access: Fully realized Parent Portal displaying student attendance metrics, grade sheets, and instant communication interfaces.

8. Conclusion

The Secondary School Portal is engineered to resolve operational bottlenecks by establishing clear role-specific workflows and ironclad data access protections. Adherence to this PRD guarantees a secure, reliable, and scalable digital foundation for modern secondary educational environments.